

SOURCE FOLLOWER

A **source follower** is a MOSFET amplifier in which the **output is taken from the source terminal**, while the input is applied at the gate. It is also called a **common-drain amplifier**.

Its main characteristic is: **Voltage gain is slightly less than 1**, but it provides **high input impedance and low output impedance**.

1. Why is it called "source follower"?

The source voltage follows the gate/input voltage:

$$V_{out} \approx V_{in} - V_{GS}$$

For small-signal variations:

$$A_v = \frac{v_{out}}{v_{in}} < 1$$

So when V_{in} increases, V_{out} also increases approximately by the same amount.

2. Voltage Gain: For an NMOS source follower with source resistance R_S :

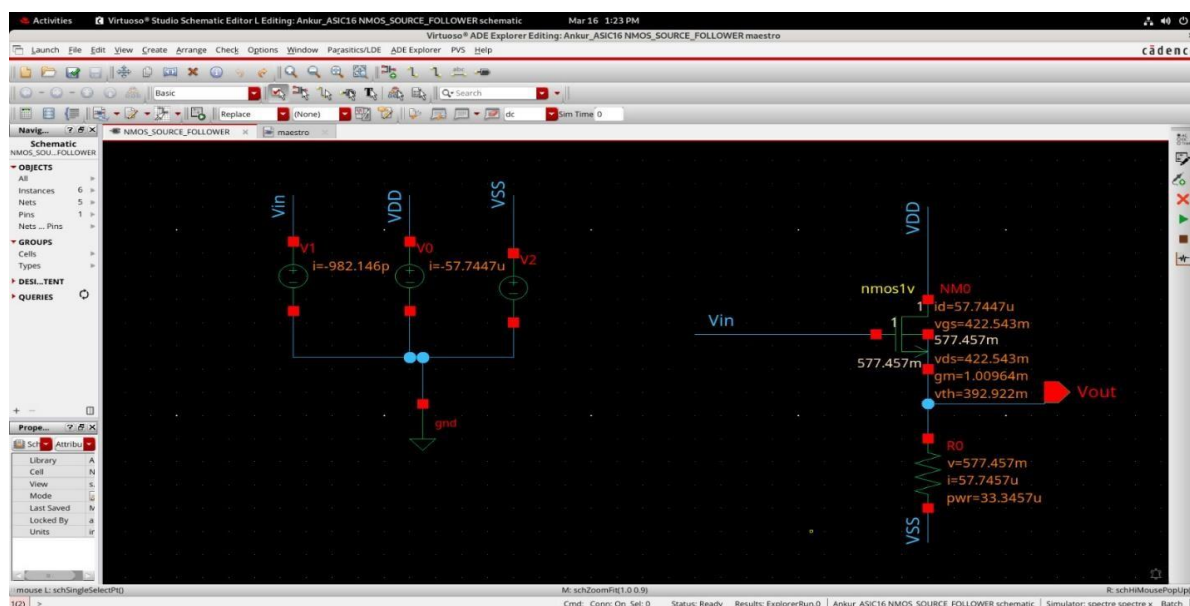
$$A_v \approx \frac{g_m R_S}{1 + g_m R_S}$$

If $g_m R_S \gg 1$

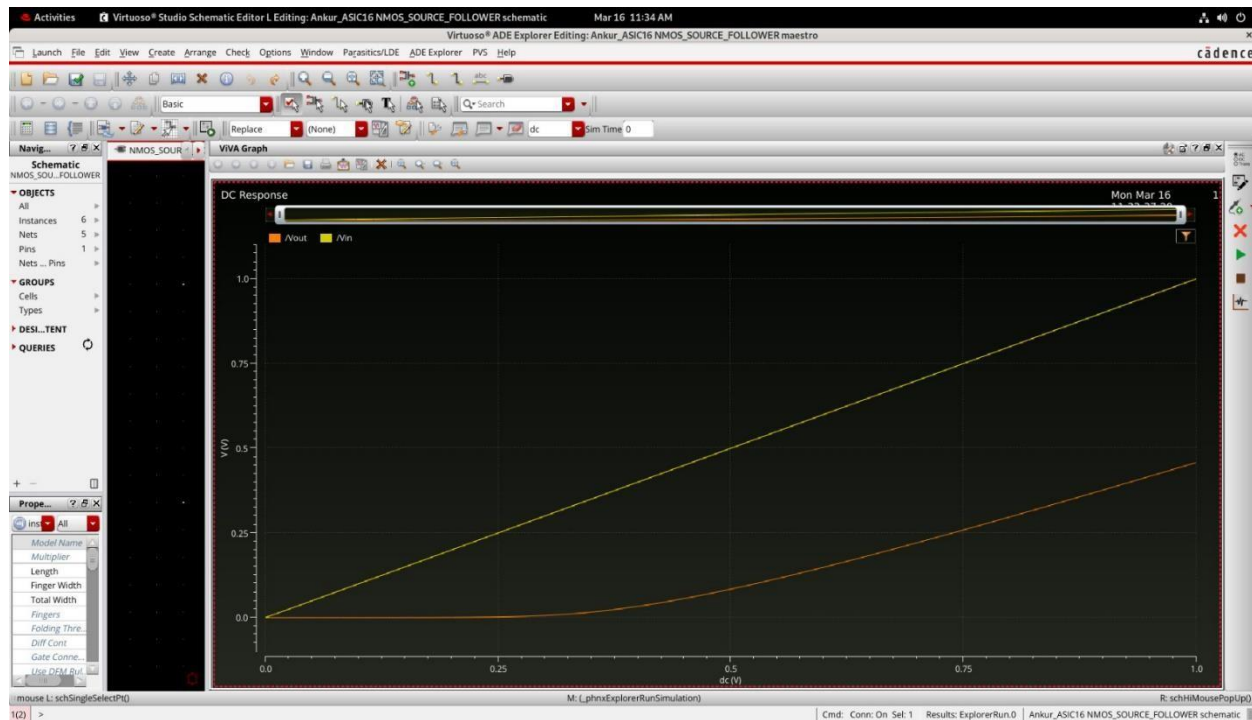
$$A_v \approx 1$$

Therefore, it is called a **voltage buffer**.

1. NMOS SOURCE FOLLOWER WITH RESISTOR



Vin vs Vout Curve For Above Circuit



Conclusion- Vout curve shifts towards Vin when width increases

Analysing the effect on various parameters related to NMOS Source Follower when width(10u) and length(1u) of device is fixed and value of resistor is varied -

R	Vth	Vgs	gm	Id	Vds	Vout
10K	396.93m	542.2m	496.20u	45.77u	542.2m	457.8m
15K	397.16m	513.04m	399.26u	32.44u	513.04m	486.95m
20K	397.33m	493.96m	336.85u	25.30u	493.96m	506.03m
25K	397.43m	480.00m	293.91u	20.79u	490.0 m	519.99m

Conclusion: On increasing the value of resistor, Vout increases since $V_{out} = I \cdot R$, Current decreases, gm decreases, Vds decreases since $V_{ds} = V_{DD} - V_{out}$ and Vout is increasing that is why Vds decreases.

Vgs decreases since $V_{gs} = V_g - V_{out} = 1 - V_{out}$ and Vout is increasing. Hence Vgs decreases.

Very slight increase in Vth value is observed.

Vout vs Vin Curve for diff values of Width



Analysing the effect on various parameters related to NMOS Source Follower when resistor value(10K) and length of device(1u) is fixed and width is varied -

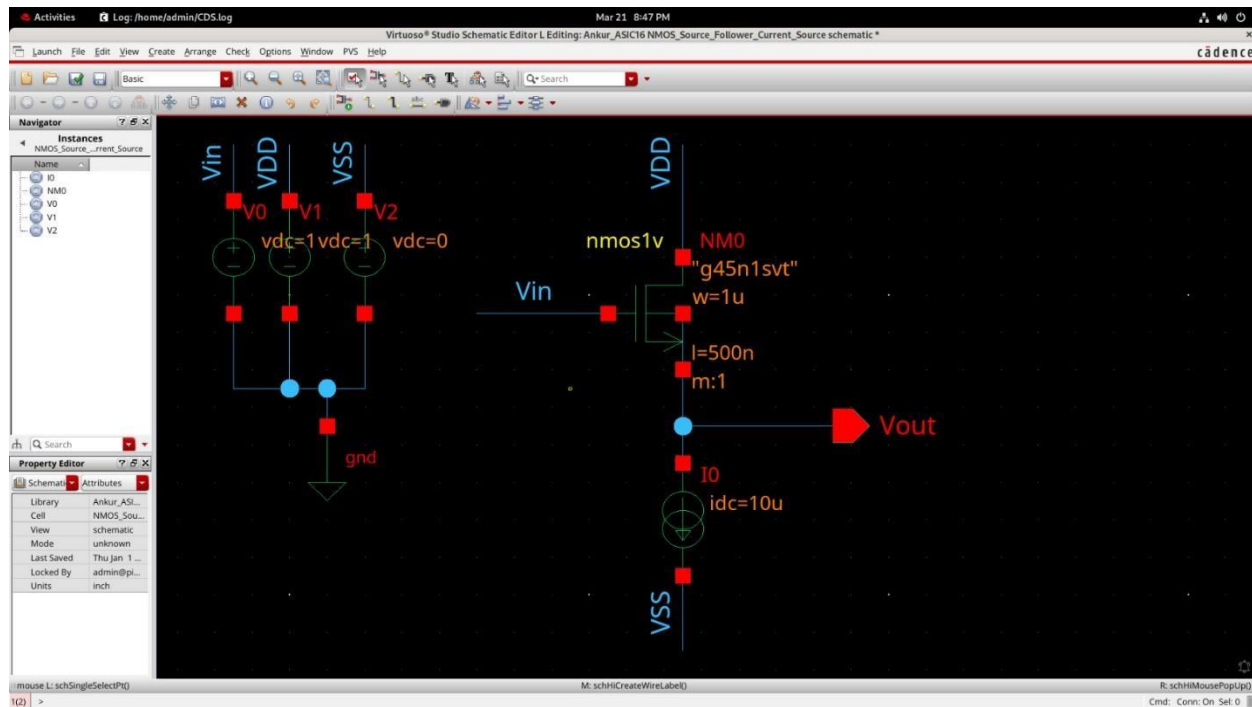
Width	Vth	Vgs	gm	Id	Vds	Vout
10u	396.93m	542.2 m	496.20u	45.77u	542.2m	457.8 m
15u	394.37m	508.005m	609.27u	49.20u	508.05m	491.99m
20u	394.90m	489.50 m	685.93u	51.07u	489.50m	510.49m
25u	393.66m	473.29 m	753.40u	52.67u	473.29m	526.7m

Conclusion: Current and gm increase as per width increases since both are proportional to width of the device at fixed length.

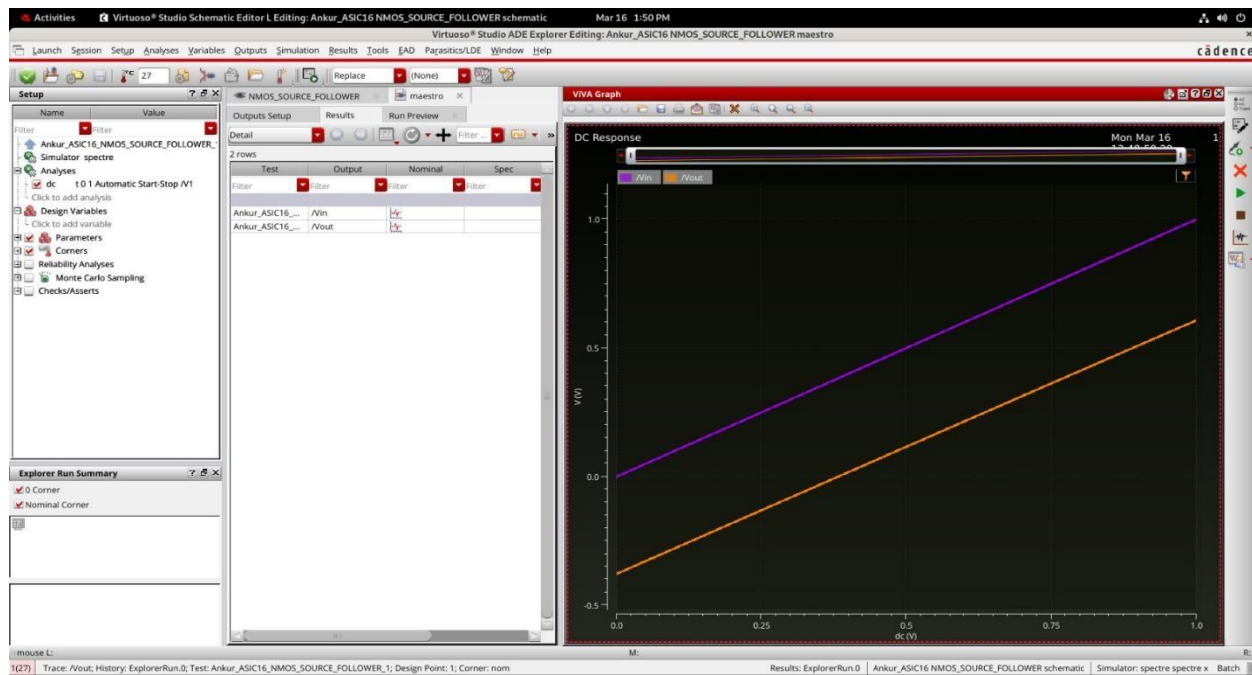
Vout is increasing since current is increasing. Hence Vgs is decreasing.

Vth is decreasing slightly.

2. NMOS SOURCE FOLLOWER WITH CONSTANT CURRENT SOURCE



V_{out} vs V_{in} curve for above circuit



Conclusion: V_{out} curve follows V_{in} with a negative offset of -0.4V.

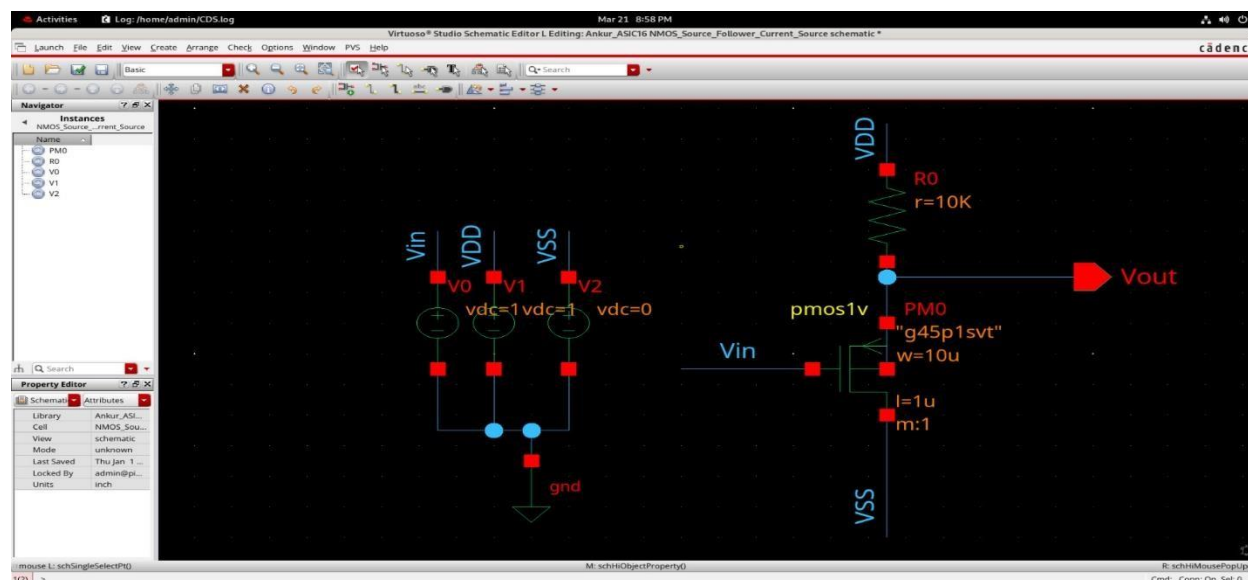
Analysing the effect on various parameters related to NMOS Source Follower with constant current source when width(20u) and length (1u) of device is fixed and value of current is varied-

I	Vgs	Vds	gm	Vth	Vout
10u	392.88m	392.88 m	196.51u	396.03 m	607.112 m
20u	429.53m	428.53 m	343.89u	395.55 m	570.46 m
30u	453.69m	453.69 m	468.68u	395.271 m	546.306 m
40u	472.3m	472.36 m	577.8u	395.07 m	527.636 m

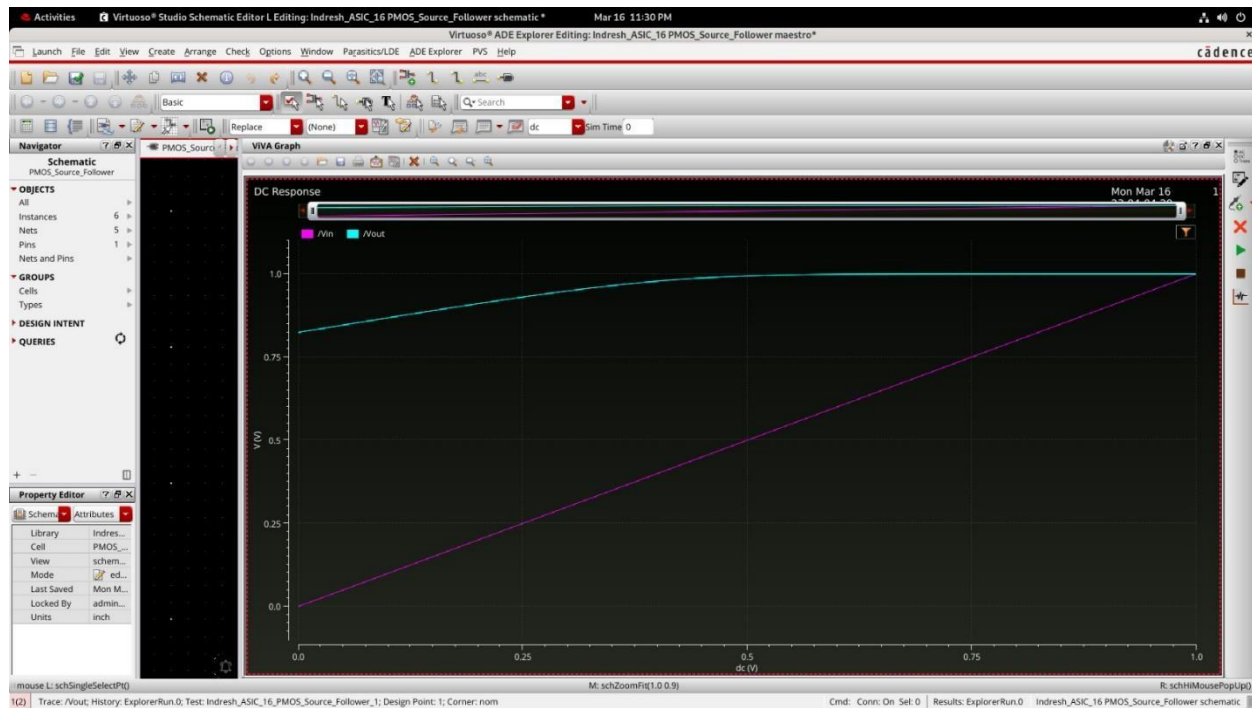
Conclusion: Since, $V_{out} = V_{in} - V_{gs} = V_{in} - (V_{th} + V_{ov})$ therefore on increasing the value of current V_{ov} increases and hence V_{out} decreases. g_m increases due to increases in V_{ov} .

V_{ds} increases since $V_{ds} = V_{DD} - V_{out}$ and V_{out} is decreasing..

3. PMOS SOURCE FOLLOWER WITH RESISTOR



Vout vs Vin Curve for above circuit



Analysing the effect on various parameters related to PMOS Source Follower when width(10u) and length(1u) of device is fixed and value of resistor is varied -

R	Id	Vgs	Vds	gm	Vth	Vr
10K	2.946p	34.64n	-1	48.47p	-561.67m	34.66n
15K	-170.19p	320.398n	-1	706.071p	-333.29m	327.80n
20K	-170.19p	429.69n	-1	706.06p	-333.29m	437.07n
25K	-170.19p	538.93n	-1	706.06p	-333.29m	546.34n

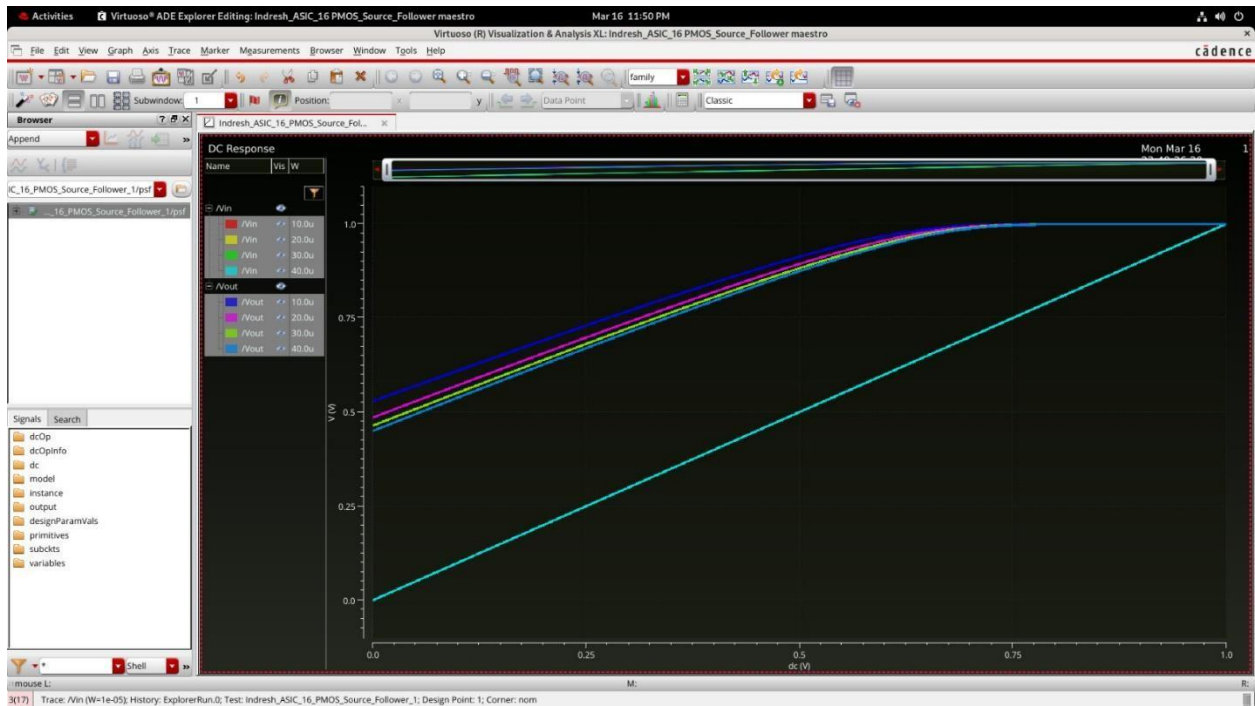
Conclusion: On increasing the value of resistor, current decreases since V_r increases and $V_s = V_{dd} - (I_d \cdot R_s) = V_{dd} - V_r$ so V_s drops and $V_{sg} = V_s - V_g$ decreases and I_d is proportional to $(V_{sg} | V_{th} |)^2$. **gm increases but it should also decrease**

V_{ds} is fixed at -1V; V_{ds} should increase but here it fixed at -1V

Since $V_{ds} = V_d - V_s$ and $V_s = V_{dd} - I_d \cdot R_s$, Here R_s increases hence V_s decreases and V_{ds} increases

V_{th} increases as per resistance increases because V_s and V_{bulk} move apart the depletion region around channel widens due to which gate must work harder to start conduction.

Vout vs Vin Curve for diff values of width



Conclusion: Vout curve shifts towards Vin when width increases.

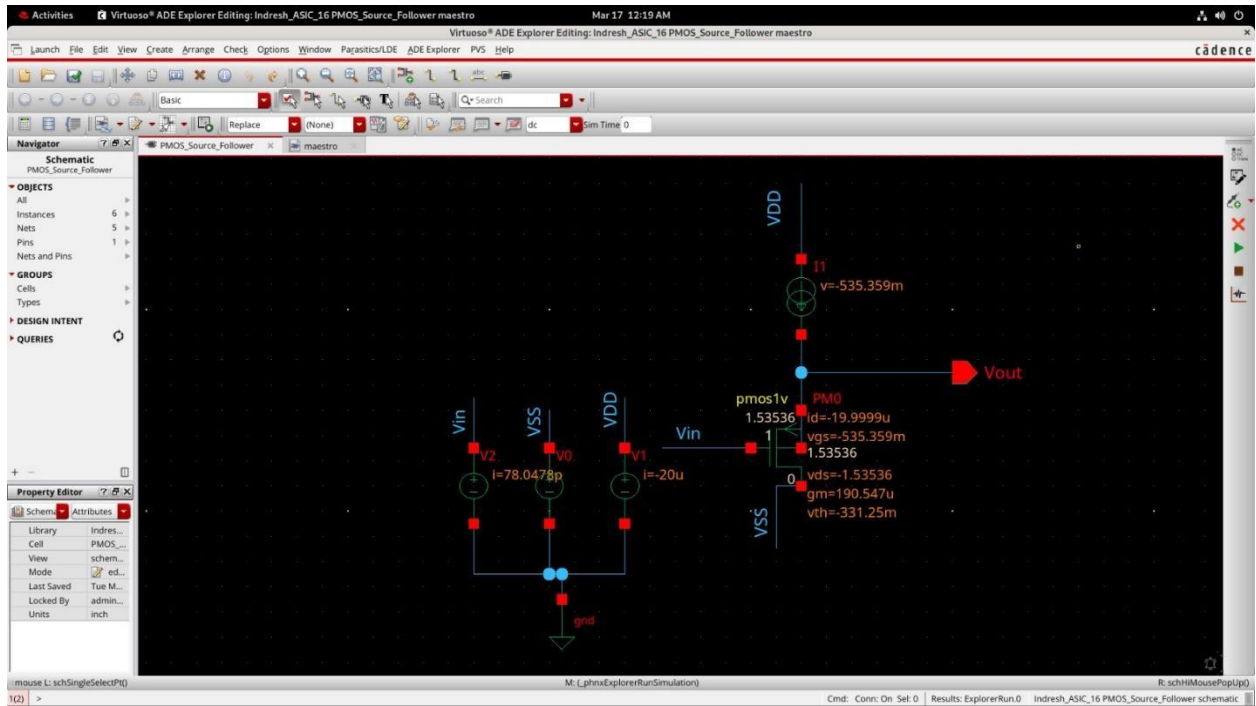
Analysing the effect on various parameters related to PMOS Source Follower when resistor value(10K) and length of device(1u) is fixed and width is varied –

Width	Vth	Vgs	gm	Id	Vds	Vr
10u	-333.29 m	211.13n	706.07p	-170.19p	-1	218.53n
15u	-333.13 m	292.59n	984.81p	-253.02p	-1	296.86n
20u	-333.30 m	380.82n	1.31n	-337.51p	-1	388.4n
25u	-333.21 m	458.85n	1.57n	-420.09p	-1	464.14n

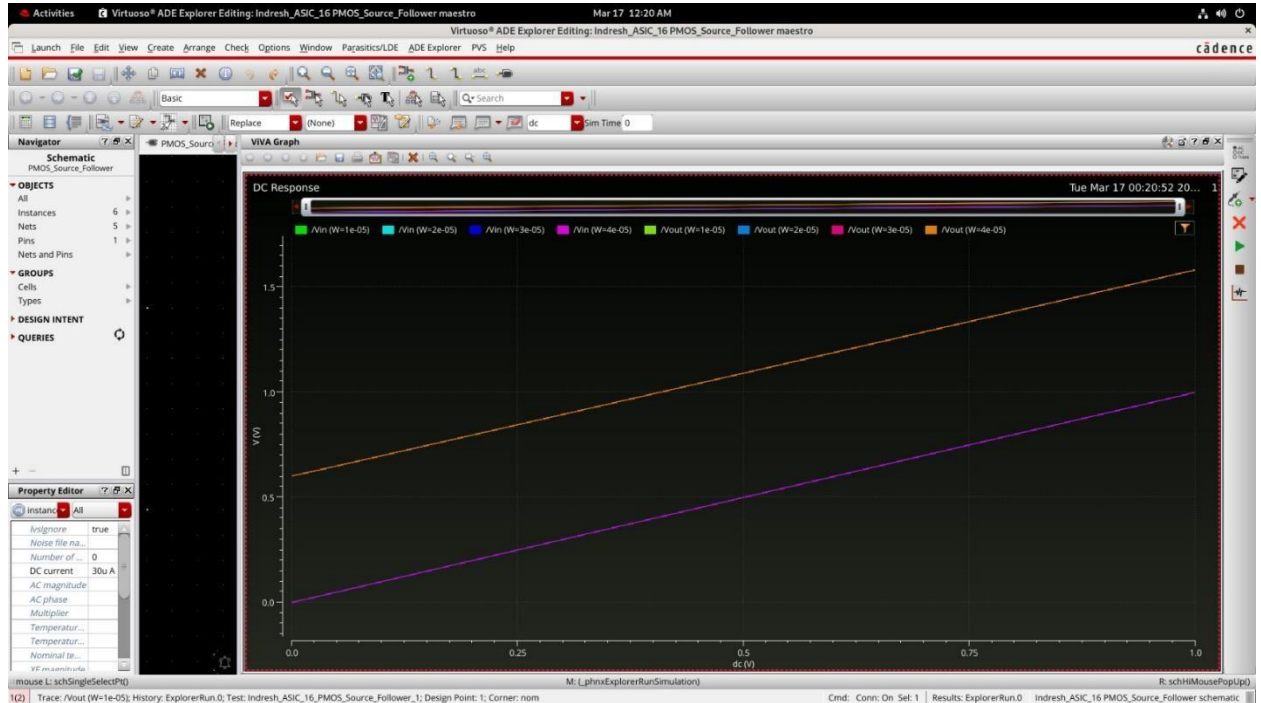
Conclusion: gm increases due to increase in width and Id also increases.

No change in Vth and Vr increases due to increase in current

4. PMOS SOURCE FOLLOWER WITH CONSTANT CURRENT SOURCE



Vout vs Vin Curve for above circuit



Analysing the effect on various parameters related to PMOS Source Follower with constant current source when width(10u) and length (1u) of device is fixed and value of current is varied -

I	Vgs	gm	Vth	Vi	Id	Vds
10u	-416.98 m	197.13u	-331.49 m	-416.98 m	-10.002u	-1.41
20u	-535.35 m	190.54u	-331.25 m	-535.35 m	-19.99u	-1.53
30u	-582.29 m	222.33u	-331.16 m	-582.92 m	-29.99u	-1.58
40u	-624.82 m	246.80u	-331.08 m	-624.82 m	-39.99u	-1.62

Conclusion: gm increases due to increase in width and current also increases.